Authentication methods in earables

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Abstract. A desire for Innovation, convenience and security has initiated a shift towards using biological data in the realm of personal authentication. This includes, but is certainly not limited to, the wealth of computing devices, such as laptops and phones, which we constantly surround ourselves with. A natural next step in this regard is to develop better authentication methods based on other physical characteristics that have displayed promise in terms of performance. Such novel techniques introduce new challenges which partially remain to be resolved, however, the popularity of ear-based media consumption devices with the ever-improving capabilities of wearable technology could offer new approaches and techniques in the future. The focal point of this paper is to introduce authentication methods that have been proposed over the last few years. Each method is centred around the collection and analysis of either brainwaves within proximity of the ear or acoustic responses from within the ear. Lastly, the methods will be evaluated and new ideas for the future will be presented.

Keywords: biometrics · earables · authentication

1 Introduction

The widespread and increasing use of phones in combination with other wearable technologies (such as smartwatches) is, regarding security, a serious cause for concern. Especially smartphones contain a wealth of personal and highly sensitive information, which in the wrong hands can cause devastating damage. This presents yet another reason for more rigorous security on these devices. Commonly however, only a simple security mechanism such as a pass code or pattern is protecting the device. Or in other cases some form of biometric authentication such as facial and or fingerprint recognition is used. These security mechanisms are a staple in the domain of biometrics. However, they are also significantly limited by their nature, as is especially the case with fingerprint recognition. To strengthen security, multiple factor authentication is commonly encouraged, especially by services with access to highly sensitive information (such as the ones in the financial or medical sphere), however adoption is usually limited, due to the inconvenience caused by employing multiple forms of authentication. This raises the question if there are secure authentication methods, which offer more security, but are additionally easy to use? The three desirable factors of authentication are knowledge, possession, and inherence. Wherein knowledge is

something known to you, such as a password or pattern. Possession is something you own such as a key, card or even phone. Inherence is some sort of uniquely identifiable biological trait (for example face, voice, eye or fingerprint). Multiple factor authentication means, multiple different pieces of information, each corresponding to a different category, must be supplied in order to verify successfully. Intuitively the more factors used, the more secure a system will be against attacks and falsification attempts. The above-mentioned popularization of wearable technologies is particularly relevant, as it enables new methods of authentication, that employ multiple factors, to be developed. The ideal system would be both secure, exhibiting multiple factors of authentication, and yet exceedingly convenient to use. Ear based authentication has been explored for several decades. For image recognition the ear is an especially viable contender because the geometry of it alters little with age and its position is generally independent of facial expression [8]. Beyond this, in recent years, further advancements have been made and novel authentication techniques with promising success postulated. These include biometric authentication based on the unique acoustic properties of the ear and in-ear electroencephalography. These two categories, which will be the matter of this report are particularly interesting as they offer several inherent advantages. (i) the nature of the systems makes reproduction and capturing attempts difficult, as there is no trace left behind (as there would be with a fingerprint). (ii) they are convenient and require little to no extra input from the user. (iii) it is possible that these systems could be readily integrated into already hugely popular earphone devices. Furthermore, their designs often already encompass multiple factors of authentication, aiding in their overall security.

2 Earables

Earables, also referred to as hearables are ear-worn devices that interact with the wearer in a multitude of ways. The most common and widespread functionality is simply audio playback and recording. New functionality and ways to interact with these devices are constantly being developed. A few methods, in regard to authentication, will be introduced in section 3 of this report. Improving battery technologies, better connectivity and smaller processors are allowing earables to perform more complex tasks, aiding in their overall usability and appeal to consumers. It is quickly self-evident by a look around that other wearable devices (wearables) have found limited adoption and are often even seen as gimmicks. In 2014, in a publishing on his website Nick Hunn argued that the market for earables may soon exceed that of other wearables such as smartwatches and wristbands [9].Within this publishing, he states, “earbuds are the only piece of wearable tech to have gained ubiquity and social acceptance”. Therefore, due to the already widespread acceptance of ear-based listening devices, further advancements and developments seem to be but a small hurdle for consumers to overcome. As more functionality and features are introduced that prove convenient and useful, it is possible to imagine that they will be easily adopted and

accepted just as wireless operation has. The ear is an excellent base to perform a variety of measurements from. It is even said to be more reliable for measurements such as heart rate or blood pressure as it is relatively stable, in comparison to the wrist, and not subject to as much movement [9]. Beyond just vital signals, future systems can be thought of that measure further biological outputs such as brain activity, stress levels, or even moods and act on these given a certain situation [5]. As is clear, the functionality imaginable is rather expansive, in this report we will be solely limiting ourselves to existing systems with methods motivated by their uses in authentication.

3 Authentication methods

Currently, there are two main types of authentication methods, based on earable technology. Namely, in-ear electroencephalography and ear acoustic feature extraction. In the following two subsections these methods will be introduced in greater detail.

Please note: references in the following section have been partially omitted to avoid strong repetition. The information within a paragraph stems from the respective paper mentioned at the start of the paragraph, unless stated otherwise or made explicitly clear by the context.

3.1 Pass-thoughts and Electroencephalography

One such authentication scheme based on electroencephalography (EEG) was proposed by Curran, Yang, Merrill and Chuang (2016) [7]. EEG is the measurement of electrical signals in the brain, stemming from the activity of the neurons. The idea of using such measurements for identification is not new, however systems proposed for EEG in the past were cumbersome, expensive, and simply not feasible. Here a more convenient, discrete, and most importantly commercially available device is used to obtain measurements. To use EEG effectively for authentication, some sort of reproducible pattern of electrical activity in the brain must be induced. This is where the concept of pass-thoughts comes in. Instead of remembering a password, the user must simply carry out a specific mental task, such as thinking of a particular instance or thing. Ideally, effective authentication will then be possible based on the collected data from the measurements. To test the viability of such a system, an experiment was performed with 12 participants. A Neurosky Mindwave Mobile wireless EEG headset was used for the measurements. Such a headset is available for 99$. It was altered slightly to make it more comfortable and effective (Figure 1). The participants were asked to perform 5 different mental tasks. Namely, relaxed breathing, imagining a chosen song, listening to a 40Hz tone, imagining a chosen face all with eyes closed respectively and lastly imagining a cube rotating with eyes open. The participants rated the tasks as easy to perform and moderately likely to use them in a real world setting for authentication. Two of the best preforming tasks (breathing

and listening) were perceived as repeatable, moderately engaging, and likeable. However, it should be noted, in order to achieve optimal results the users would most likely not be able to choose their task. The obtained data was analysed in two ways, within- and between-participant analysis. Within-participant analysis was to evaluate how well authentication would perform, using different tasks of one participant. Between-participant analysis was to evaluate how well authentication would perform using all 5 tasks from all 12 participants. Ultimately, for within-participant analysis using the best performing task for each individual an authentication accuracy of 71.9% was achieved. For between-participant analysis, again using the best performing task for every individual, this was slightly higher at 80%. Overall it was shown that a simple, low-cost consumer grade device can be used to attain a decent degree of authentication accuracy, however, as is also stated in the conclusion "the achieved rates are not immediately viable for real-world authentication applications" there are still improvement necessary before the proposed system is practicable.

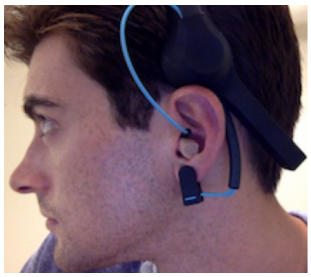

Fig. 1. Modified Neurosky MindWave headset used by Curran et al [7].

A different paper by Curran, Merrill, Chuang and Gandhi (2017) [6] also presents an EEG based authentication system, although with more accurate hardware and emphasis placed on other features. Convenience and usability were prioritized over cost. Instead of opting for a commercially available solution, a highly specialized device was used to take EEG measurements (see Figure 2). For each user, an earpiece must be produced individually. Meaning, a mould in addition to a 3D scan of the ear must be taken, with which the earpiece can then be produced. 3 Silver chloride electrodes are inside the earpiece, these perform the EEG measurements. The uniqueness of the earpiece aids in security because it can act as a third factor of authentication. It is considered the physical token, much like a key or card would be. The uniqueness of an individual's brainwaves (measured through EEG) satisfies the inherence factor and the use of a passthought (a mental task instead of a password) satisfies the knowledge factor. Therefore, this system, based on a special earpiece can achieve all 3 factors of authentication. To test the proposed system an experiment with 7 participants was performed. Each participant wore their individual earpiece, in addition to another electrode place above the left eye. This extra electrode is not part of the proposed system and is simply used as a control mechanism. As in the other

paper by Curran et al. (2016) [7] a series of mental tasks were chosen, with the additional goal of being as diverse as possible. EEG measurements would then be taken, while the participants performed the mental tasks. The data collected substantiated the accuracy of the ear-based measurements, as they matched the readings from the scalp electrode. To evaluate the results the false rejection rate (FRR) and false acceptance rate (FAR) was calculated. A false rejection is when the system incorrectly rejects an input (rejects an authorized user). A false acceptance is when the system incorrectly accepts an input (accepts an unauthorized user). Each participant had at least 1 task for which both the FAR and FRR were 0%, overall the worst FRR was 6% and the worst FAR was 1%. Across all tasks the mean FAR and FRR were both less than 1%. Furthermore, the correct participant performing the wrong task would always be rejected. Earpieces were also swapped (in a single case between two participants) and no successful authentication attempts could be made with the swapped earpiece. Generally speaking, The results and findings portray this sort of system as feasible.

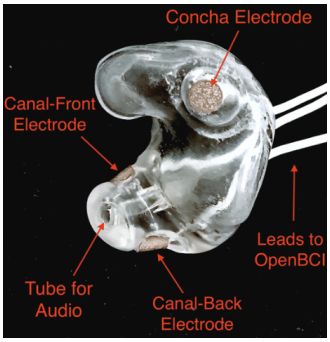

Fig. 2. Depiction of the custom earpiece used by Curran et al. [6].

Developing a novel EEG authentication system which can produce results on par with conventional and unwieldy equipment was also the goal in the paper by Nakamura, Goverdovsky and Mandic (2018) [12]. The system had to be wearable, easy to use and fast to setup, yet still be able to produce accurate results. Ultimately an expanding foam earplug was used, which had two flexible electrodes attached to it. The foam allows the contraption to snugly sit at the ear canal opening, pressing the flexible electrodes against the top and bottom of the ear canal. Additionally, a ground wire was placed on the lobe. Using such a recording device is advantageous as it can fit many different ear shapes, due to its workable and soft material. To investigate the results achievable with this system recordings were conducted with a group of 20 individuals. Every participant wore the device in their left ear. Multiple Recordings were taken during two different sessions, separated by an interval of 5 to 15 days. During the recordings, the participants where simply asked to rest with eyes closed for 190 seconds, no other mental tasks had to be performed. The space between the sessions was important to introduce variability, in order to better simulate real

world usage. The obtained double channel EEG (one from each electrode) can then be preprocessed and further analysed. To be able evaluate the feasibility of the system, the recordings from the two days were explicitly split. Data from the first day of recording was used as training data for a classifier. The recordings from the second day were then used as a validation set to evaluate the accuracy of the classifier. The original 190 second recording obtained was split into different segments of 10, 20, 30, 60 or 90 seconds. For each of the segment sizes a separate evaluation was done. It was found that the 60 and 90 second segment sizes produced similar results and performed the best out of the segment sizes. Specifically, with a segment size of 60 seconds an FAR of 2.3%, FRR of 32.2% and an authentication accuracy of 95.7% were achieved. The overall trend seen is that longer segment sizes allow for greater accuracy. The 90 second segments did perform slightly worse than the 60 second segments, it is suggested that the declining number of segments with increasing segment size might not be sufficient training data. Naturally, longer segment size would also mean longer recording time, which is not optimal for an EEG biometric authentication system.

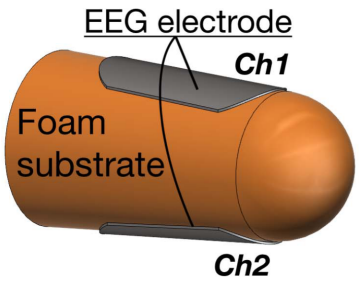

Fig. 3. The foam piece, with two attached electrodes used by Nakamura et al. [12].

3.2 Audible and inaudible stimuli

In a paper by Mohammed Derawi (2016) [8] an intuitive, simple system to measure acoustic properties of the ear is introduced. It consists of headphones that surround the ear, an amplifier and a laptop (complete with an assisting software). The headphones have been slightly modified; a hole has been drilled through the outer shell of one of the headphone cups (4). The headphones and the microphone are both connected to the laptop. This allows the laptop to both generate and record the reverberation of the signals, through the headphones and microphone-amplifier pair respectively. Once the emitted signal has been collected, specific acoustic features can be extracted from it. The idea is that the unique individual geometry of the outer ear (ear canal and pinna) will dictate a unique echo of this signal. This unique echo can then be used to verify an identity, by comparing it with older recordings and calculating a similarity. Solely using the echo

alone for this purpose is not sufficient. The collected signal must undergo preprocessing, before it is usable. The goal of the preprocessing is to eliminate the influence of the microphone and the headphones on the signal, so ideally only the interaction between the ear and the signal remains. Removing the interference and unwanted effects of other objects (such as the recording apparatus) from the echo, will allow for a cleaner extraction, meaning ultimately a more distinct classification. A study with 50 candidates was performed, for each 50 measurements were taken. The candidates were instructed to remove the headphones between every measurement. Sets of features were extracted from the collected samples and classified. The best verification performance achieved was with an EER of 14.9%. Furthermore, the random forest classifier correctly identified 74.36% of the instances. While these rates are not sufficient for meaningful authentication, they do display the effectiveness a rudimentary system consisting of only commercially available components can achieve.

Fig. 4. Microphone-integrated earphones used by Mohammed Derawi [8].

In their paper Arakawa et al. (2016) propose a similar approach [3]. The focal point is also authentication based on the acoustic properties of the ear, however, not the entire outer ear, but only the ear canal. For this method, only the echo within the ear canal is important. To achieve isolation of the ear canal, instead of large headphones, specialized earphones with an integrated microphone were used. The microphone earphone combination was handcrafted. Furthermore, earphones were chosen over other audio devices with the intention of further reducing within-individual variability, stemming from the high degree of physical freedom some devices allow while being worn. The speaker on the earphones emits a signal, the microphone can then pick up the echo of this signal and lastly the recording can be further processed. From the recorded echo, further information is derived. This information is used as a feature representation, meaning rather than using the entire response signal for analysis only certain components are extracted. These specific feature representations have been found to be particularly effective in other fields (such as speech and emotion recognition). Additionally, they are easier to work with and compute. Through these extracted features, a similarity can be determined and used to authenticate an individual. Keeping computation complexity low was an additional goal, to al-

low the proposed system to work seamlessly on a phone. As a result, similarity measures were kept simple. Using the aforementioned device and methods, an experiment was preformed and acoustic data was collected from 45 individuals. Fortunately, due to the nature of the earphones, noise from outside the ear is muffled. Noise stemming from the inside, such as the heartbeat, breathing, muscle movements and vocal sounds had to be filtered out. Three different feature representations were used and compared to each other. By far the Mel-Frequency Cepstral Coefficients feature representation outperformed both the other representations in both similarity measures achieving a EER of 0.97%. Interestingly, a working prototype for an Android app is also introduced. Through which users can simply register by having the acoustic features of their ear captured. Afterwards the system can authenticate them, by comparing a new signal, with the ones recorded at registration. However a specialized earphone with an embedded microphone is needed for this to work (see figure 5).

Fig. 5. Earphones with embedded microphone from NAP enterprises [13].

Building on these ideas, a paper by Mahto, Arakawa and Koshinak (2018) introduces the novel concept of authentication with both inaudible and audible signals [11]. The use of audible signals may create significant discomfort, the sound could be considered disruptive and disturbing by the user. Naturally, inaudible signals are advantageous as they cannot be perceived, therefore, authentication can be invoked frequently without being an inconvenience. Despite using inaudible signals, the fundamentals of this system are similar to the ones using audible signals, acoustic properties of the ear canal are extracted and then authentication can be performed with this data. For each authentication trial a probe signal is transmitted and recorded. This is done with a pair of microphone integrated EarPods capable of frequencies up to 48KHz. The frequency hearing range in adults is usually around 16Hz - 20KHz. Therefore, to achieve silent authentication a frequency above 18KHz is used. This frequency was found to be especially suited for capturing individual acoustic features (representing person dependent traits). The high frequency short wavelength is proficient at measuring certain properties of the ear canal such as length and curvature, as even the smallest variations can be vital for authentication. Lastly, high frequency signals are particularly robust against background noise and music, meaning they

can be effectively used for authentication of a user, even while other sounds are present. The recorded echo can then be compared to an echo obtained during an enrolment stage. For each of the echoes, acoustic features can be extracted using among others, Mel-Frequency Cepstral Coefficients. Cosine similarity was chosen as a similarity measure. Meaning, if the signals features show a satisfying degree of similarity in the cosine measure, the user can be authenticated. In the evaluation performed, it was found that, inaudible signals are significantly effected by the relative position of the earphone against the ear canal. Over multiple trials, where the relative position was adjusted, the measurements altered drastically. This means the measured acoustic features showed clear deviation. It is suggested that the smaller wavelengths are simply more sensitive to the relative position of the earphone, thus creating a greater discrepancy when the position is altered. The within-individual variability resulting from relative position of the earphone is also present with audible signals, however, to a lesser degree. Therefore, it is suggested that an optimal system may use audible signals for initial authentication of a user. Initial authentication is simply a cold start, meaning when the user needs to be authenticated and has not done so in the current wearing session (for example the first time an authentication request is made, every time the device is taken off an put back on again). When the user has been initially authenticated and the relative position is no longer expected to change until the device is taken off, inaudible signals can be used for further authentication requests. This is known as continuous authentication. In the experiment performed on 25 subjects, the EERs for continuous authentication are less than 0.01% for both audible and inaudible signals. For initial authentication, the best EER achieved was 1.44% with audible signals. The combination of audible and inaudible signals seems to be an effective approach in providing consistent, but also convenient authentication with earables.

A moderately different approach is presented in a paper by Liu and Hatzinakos [10]. Wherein the acoustics of the ear are still important, but instead of capturing the echo from a known signal, another sort of response is captured and used for authentication. It has been called transient evoked otoacoustic emission (TEOAE for short). TEOAE refers to a feint acoustic response from the cochlea (a part of the inner ear), that is approximately 20ms in length and can be triggered through a low level click sound. Within the cochlea, the outer hair cells act as an amplifier for incoming sound by vibrating. This movement of the outer hair cells produces a nearly inaudible sound known as otoacoustic emission (OAE). This OAE then echoes back through the ear canal where it can be measured by a microphone and used for further analysis. The raw recorded response is not suitable for meaningful authentication. Due to how TEOAE works, different frequency components of the response arrive in the ear canal at different times. In this regard it was found most effective to extract particular individual frequency components, that are well suited for differentiation. The response from two different subjects might look similar at first, however the extracted component could be significantly different. Additionally, results from a separate recording

sessions of an individual might differ slightly, but the extracted components will display high degrees of similarity (see Figure 6).

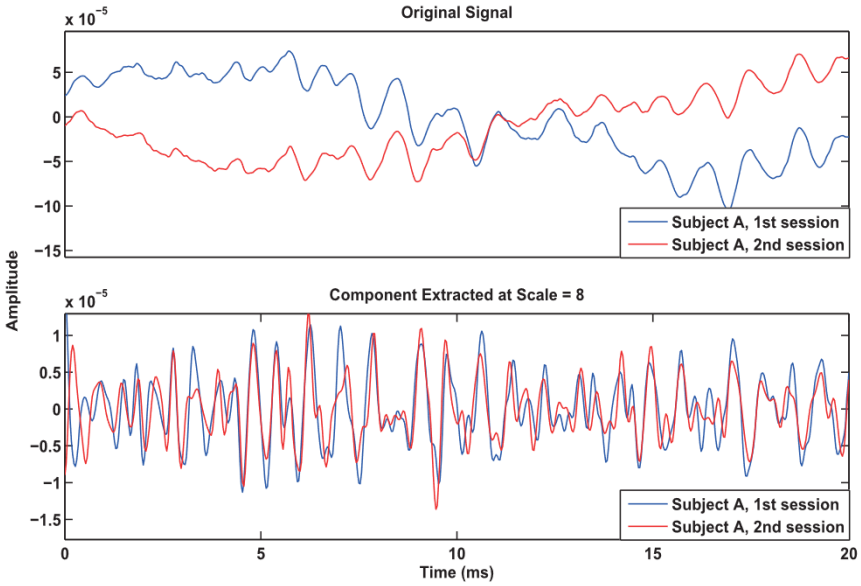

Fig. 6. *Top*: TEOAE captured from an indivudual across two recording sessions. *Bottom*: Extracted frequency components of the captured emissions[10].

The processing of a signal allows for low computational complexity and resource requirements, additionally it also helps reduce within individual variability. An experiment was preformed to evaluate the proposed methodology. A vivosonic integrity system was used for measurements, although it was also suggested that a simple microphone integrated earphone or alternatively a small probe would be sufficient for the purposes. Recordings were done on 54 subjects in a regular office. TEOAE responses from both the left and right ear were collected during two sessions, with a gap of at least one week in between. For verification EERs of as low as 0.39% were achieved in the right ear. For identification rates of as high as 96.3% where achieved, also in the right ear. The performance of the left ear was generally slightly worse than that of the right ear, however, this was not surprising as biological studies have found the TEOAE response to be more significant from the right ear [14]. When considering both ears simultaneously, various fusion strategies to combine data from the left and right ear can be used. Ultimately, resulting in an identification accuracy of greater than 99% and also improving upon the single ear performance in verification. Overall, the TEOAE approach showed favourable performance. Considering its other qualities such

as universality, uniqueness and stability it can definitely be considered a feasible biometric modality.

4 Discussion

Table 1 and 2 contain a performance overview of different authentication systems introduced in section 3. It is crucial to note here, that the results must be surveyed cautiously as each system was evaluated differently. In some cases, the entire methodology of the evaluation and data collection varied drastically. Some evaluations were performed with as little as 7 participants [6], others with as many as 54 [10]. Furthermore, for some evaluations data was collected across multiple days, for others there was only a single recording session. Additionally, how the training and validation sets were constructed would usually depend on what produced the best results, thus the training and validation split as well as whether or not the sets were disjoint, was not uniform. Naturally, as is highlighted by Liu and Hatzinakos [10], this can introduce all sorts of biases and lead to results that are not representative of the real world. Assuming for example in the case of EEG signals, that these vary from day to day, alter with time within an individual or even depend on mood or stress levels. As a result of this, collecting data from an individual only on a single day during a single session, where the signal might display strong consistency is not optimal. An evaluation of the system, with data collected in this manner, will not give an accurate representation of how well the system may function because a potentially great source of variability and difficulty is simply avoided. Lastly, the figures represented in 1 and 2 are usually the highest stated within the respective papers, thus it is not advisable to compare these systems solely through the values in the tables.

Table 1. Overview of acoustic authentication performance.

	FAR	FRR	Identification	EER
Outer/Inner ear [8]	-	-	74.26 %	14.9%
Ear Canal [3]	-	-	-	0.97%
Inaudible/continuous [11]	-	-	-	1.44%
TEOAE [10]	0%	1.55%	99.4%	0.02%

Table 2. Overview of EEG performance.

	FAR	FRR	Identification	EER
Specialized earpiece [6]	<1%	<1%	100%	-
Commercial headset [7]	0.258%	0.199%	80%	-
Foam/electrodes [12]	2.3%	32.2%	95.7%	-

How do the systems compare to exisiting ones? It is difficult to find publicly available information about the FAR, FRR and identification accuracy of biometric modalities that are currently commonly used and implemented. Especially, fingerprint and facial recognition would be interesting as a comparison, due to their widespread usage on smartphones. Limiting ourselves to the Face ID and Touch ID systems from Apple, it is stated by the company on their website that the probability of a random person being able to match a single enrolled fingerprint is 1 in 50000 [2]. Respectively, for Face ID this same scenario has a 1 in 1000000 probability of occurring [1]. In this regard it is difficult to pass any sort of judgement on the systems introduced in section 3.

comparison by other metrics In a paper by the German government the basic requirements a biometric modality should fulfil are highlighted [4]. The underlying trait or characteristic used by the system must be universal, meaning every person has it. It should be unique, meaning that no two people should share it. Lastly, the property should be easily measurable (collectability). There are further aspects that are not absolutely necessary but are still significant as they would increase the overall quality of a system. These are factors such as price, acceptance by the population, ease-of-use, and performance. The systems introduced in section 3 display the collectability of EEG signals. Three different approaches and also devices were used to extract EEG signals. For each of them ease of use, overall convenience and performance were all different. Furthermore, EEG signals are by nature universal and can (as seen in table 2) be used to discriminate between individuals effectively, thereby exhibiting uniqueness. Overall, the true feasibility of all three EEG systems in their current state is uncertain. The solution using a commercial headset could still be seen as too cumbersome and especially non discrete, possibly limiting its acceptance and usability. Using custom fit earpieces produced excellent results, however making such an earpiece is certainly costly, hurting the economic viability. Simply using a foam earplug seemed to be a very intuitive approach which also performs decently well and is far simpler than a custom-made earpiece. Nevertheless, the FRR observed could possibly lead to frustration and once again hurt its usability. Additionally, a foam earpiece is a great leap away from the earphones the public is so familiar with, casting further doubt regarding adoption. Similarly, with authentication based on the acoustics of the ear, universality is once again inherent. TEOAE is somewhat special in this regard but was also found to be present in upwards of 99% of all ears. This is of course not entirely universal but can most likely be considered sufficient. TEOAE as well as other ear acoustic characteristics such as the ear canal transfer function have been found to be highly individual, thus fulfilling uniqueness. Lastly, collectability is met by the success of the methods introduced in section 3. The acoustic based authentication systems from section 3 have an inherent advantage over the ones using EEG. This is because most of the devices used to extract the acoustic characteristics, were either readily available earphones, or slightly to heavily modified custom ones. The devices must be able to record and play an acoustic stimulus, fortunately regular earphones that are very widely spread today, are already capable of media playback. Therefore,

a possible integration of one of the acoustic systems with existing technology would be far simpler and probably also met with less scepticism. This is backed up by the existence of purchasable earphones with in-ear recording capability 5. Conversely, integrating an EEG measuring device into an earphone, was shown to be possible [6], however, it comes with added complexity. As mentioned earlier, earphones for listening to music are currently the only truly socially acceptable earable, any system not able to provide a media playback feature will probably only achieve limited adoption. Despite this both EEG and acoustic authentication have an advantage over existing biometric modalities such as the fingerprint as it is extremely difficult to capture and reproduce either of these signals meaning attempted access though forgery or spoofing would prove to be difficult.

5 Future work / ideas

The introduced methods were certainly able to show early signs of promise. For continued research in these matters the effect of a larger sample size will have to be unquestionably considered and investigated. All the papers featured a relatively small sample size of participants for their evaluations. This is understandable as the focal point of the papers was not to show viability of the entire system, but rather to act as a proof of concept introducing new ideas as well as techniques to collect biometric outputs. Therefore, larger scale trials would certainly be helpful to further assess the capabilities of the systems, which already preformed considerably well in their original tests. Will the performance be unaffected? Will the system be able to discriminate between individuals with possibly similar but not identical traits? Will effective identification of individuals within a larger group be possible? These questions and more will hopefully be answered with more research. Regarding the evaluation additional adjustments should also be made in the future to reflect a more accurate use case. If measurements can be collected during multiple adequately spaced-out sessions, at different times of the day, this will give greater insight into the true feasibility of the authentication system. The permanence of a characteristic and its stability are crucial aspects, that might not be represented sufficiently in only a single recording session. Furthermore, the influence of physical changes within the ear, such as wax build-up, would be another interesting factor to explore, especially in how it would, if at all, change the acoustic properties of the ear. Additionally, the environment wherein the recordings were taken should also be expanded upon for future experimentation. Typically, this was done in a relatively quiet and calm office or lab space. Including louder environments like a restaurant, shop, park, or a busy street would possibly introduce greater variability and noise in the measurements. This would be helpful to discern possible limits of an authentication method. Lastly, in their paper Liu and Hatzinakos [10] displayed, that by using measurements from both ears as opposed to just one they could improve the performance of their system further. This suggests that if it is possible to design an earable with both EEG and acoustic capabilities it could possibly perform better than any system based on a single characteris-

tic. Furthermore, the acoustic systems displayed remarkable differences in what they measured and quantified. TEOAE and more regular ear canal acoustics for example, both rely on the acoustic properties of the ear, yet could be considered different biometric characteristics. Nevertheless, the recording devices for both make use of a similar principle, therefore offering another promising opportunity for possible future combination.

6 Conclusion

In this report, seven novel authentication systems based on two biometric modalities (EEG and ear acoustics) were introduced. Overall, the systems had varying performance, but even the most rudimentary can be considered a proof of concept for the respective underlying biometric characteristic. It is interesting to investigate new ways of authentication because current methods such as fingerprint recognition are prone to falsification and attacks, something EEG and ear acoustics would be naturally more resilient against. It is well established that EEG and acoustics of the ear display desirable qualities as authentication methods, however, collectability has presented itself as a major challenge so far. This challenge was approached and partially resolved in the introduced papers, instilling confidence that new improved biometric systems are a realistic goal for the future. Confidence exists primarily in the possible integration of the authentication methods out of section 3 with regular earphones, providing a new avenue of discrete and convenient authentication to a user.

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